

Exercise 04: Time Response

0. Block Diagrams

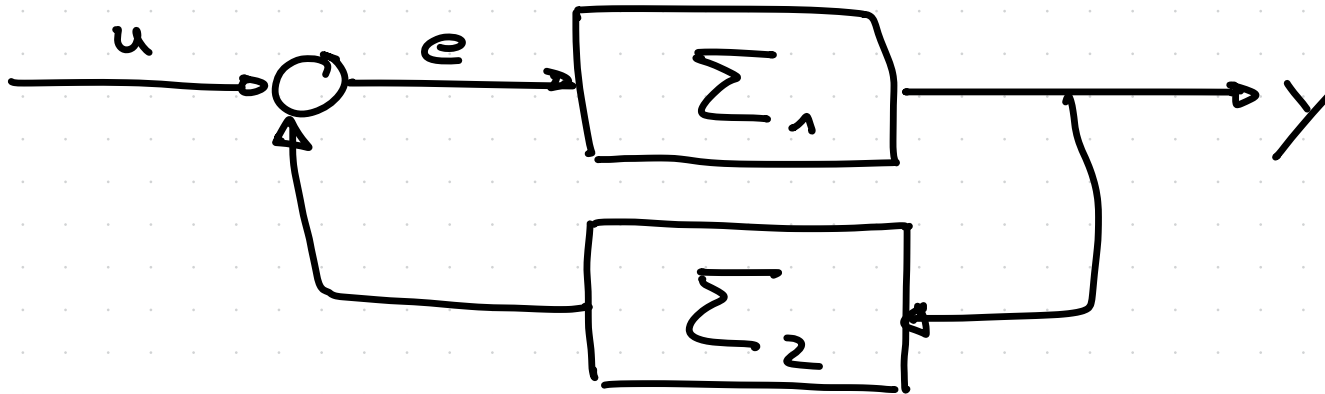
1. Introduction / Orientation

2. Time Response of a scalar LTI-System

3. Generalization to a higher-order System

4. Stability

0. Block Diagrams (PSO1, Ex 1)



$$y = \Sigma_1 u$$

$$e = u - \Sigma_2 y$$

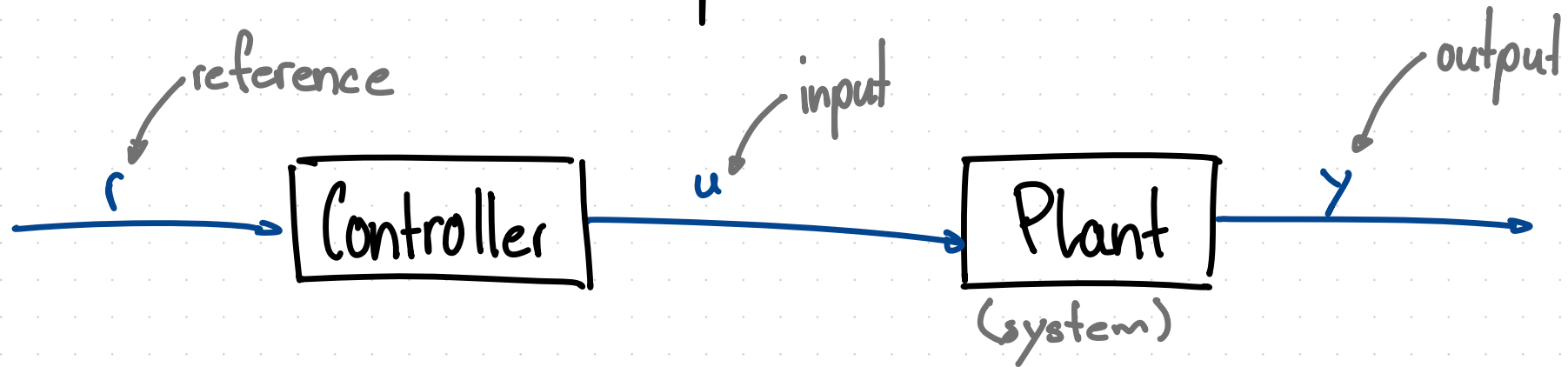
$$\rightarrow y = (I + \Sigma_1 \Sigma_2)^{-1} \Sigma_1 u$$

$$\rightarrow y = S_1 (I + S_1 S_2)^{-1} u$$



Where we left off...

- General structure of our problem:

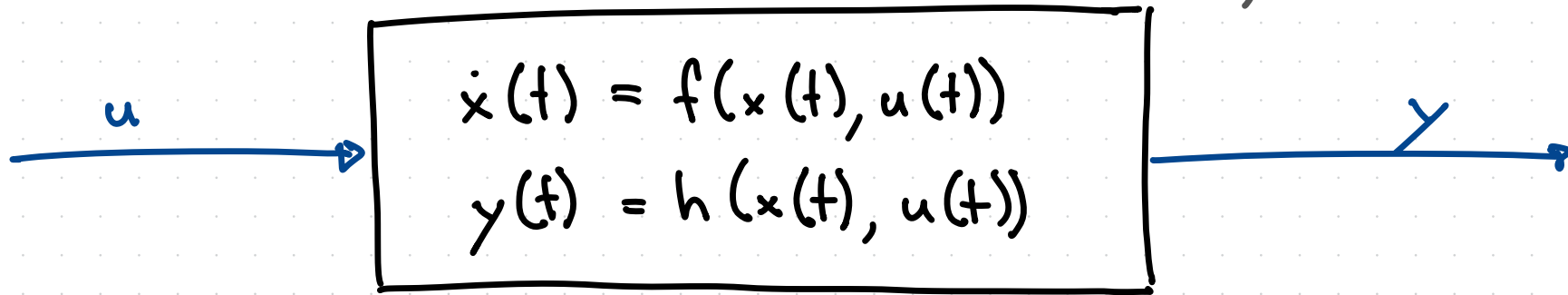


- Focus on our general system:



• Which is really just...

Mathematical formulation
of our system



↳ we can find $f(x(t), u(t))$ & $h(x(t), u(t))$ through modeling.

BUT... In general this is hard to control

• We can transform any system into an LTI-system through **Linearization**

- ↳ choose an equilibrium point
- ↳ apply Taylor series

$\Rightarrow$

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

→ Instead of complex equations, we can now characterize the system's dynamics through A, B, C, D and the initial condition x_0 .

2. Time-response of a scalar LTI-System

• Problem statement: *fully describe our system*

Given: A, B, C, D

$$x(0) = x_0$$

$$u(t)$$

Find: $x(t), y(t)$

• We will first try and find a solution for the simplest setup:

- first order sys.
 - Single Input
 - Single Output
- } $\Rightarrow A, B, C, D$ become scalar:
 a, b, c, d

$$\dot{x}(t) = ax(t) + bu(t)$$

$$y(t) = cx(t) + du(t)$$

$$x(0) = x_0$$

What now?

- What do we know about our system?

- ↳ Linear, Time-invariant & Causal

- Causality $\Rightarrow$ Past summarized in state $x(T)$

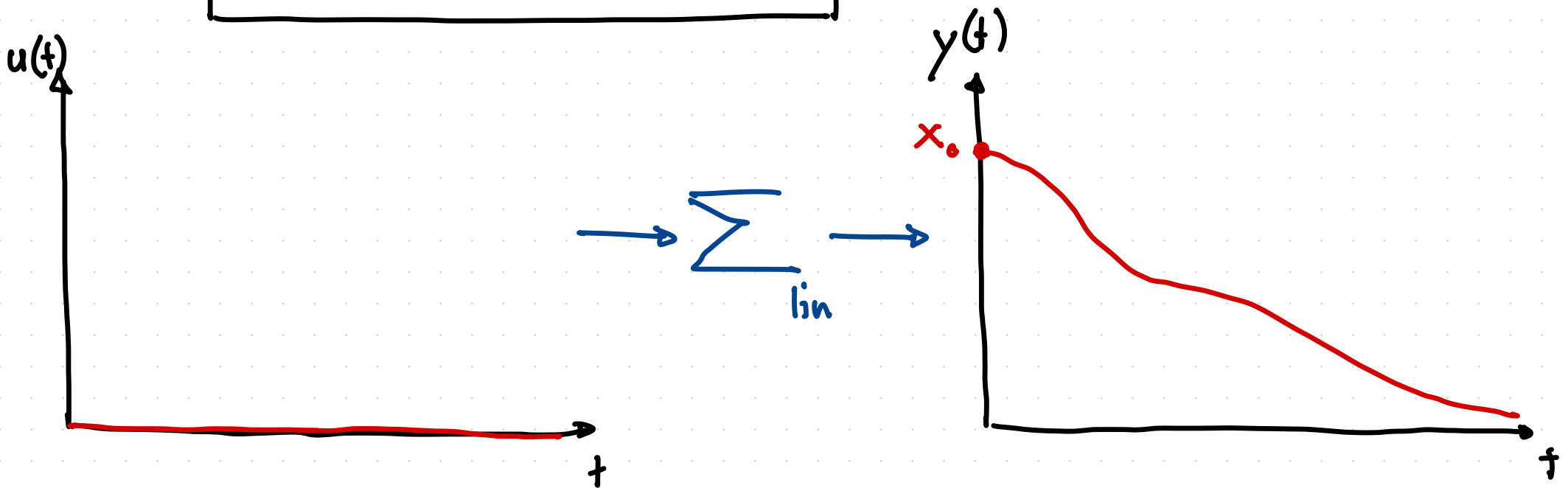
- Time-invariance $\Rightarrow$ Time reference does not matter
 $\Rightarrow$ We can choose $T=0$

• We will describe $y(t)$ using 2 terms:

i) Initial Condition response y_{ic}

$$\begin{aligned}\dot{x}(t) &= ax(t) + \cancel{bu(t)} \\ y(t) &= cx(t) + \cancel{du(t)} \\ x(0) &= x_0\end{aligned}$$

$\Rightarrow$ Set $u(t) = 0$



ii) Forced Response

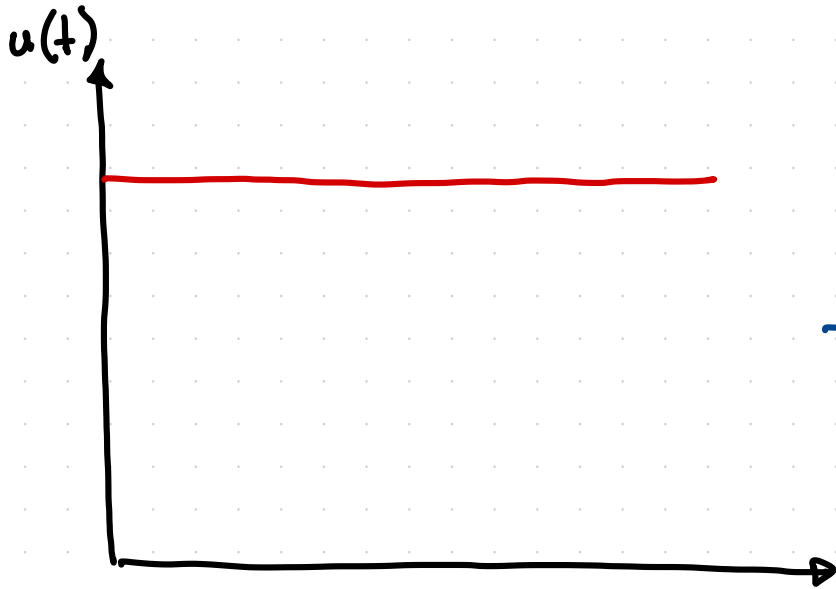
y_F

$$\dot{x}(t) = ax(t) + bu(t)$$

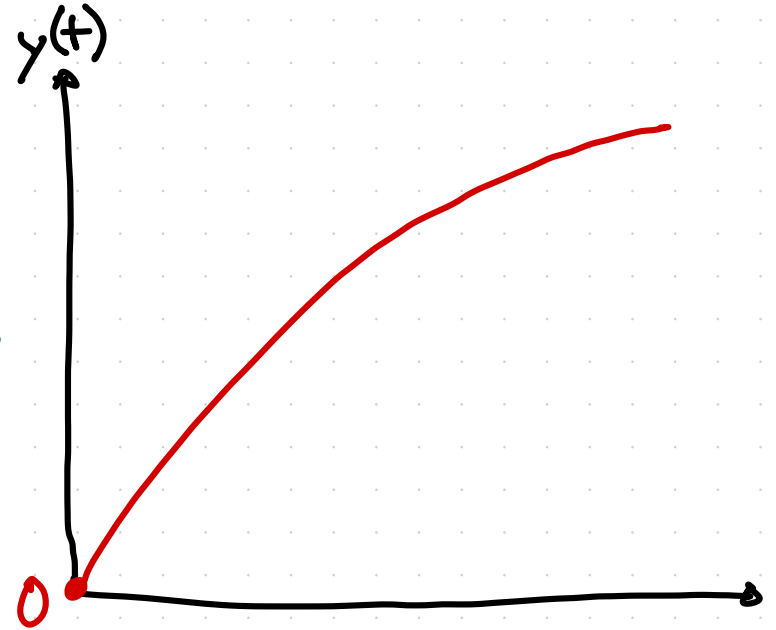
$$y(t) = cx(t) + du(t)$$

$$x(0) = \cancel{x_0}$$

$$\Rightarrow \text{Set } x(0) = x_0 = 0$$



$\rightarrow \Sigma_{\text{lin}} \rightarrow$



$\Rightarrow$ Express y through y_{ic} & y_F :

$$y = y_{ic} + y_F$$

i) Initial Condition Response

• Problem setup:

$$\dot{x}(t) = ax(t) \quad (\text{I})$$

$$y(t) = cx(t) \quad (\text{II})$$

$$x(0) = x_0 \quad (\text{III})$$

$$(\text{I}): \frac{d}{dt} x(t) = a x(t)$$

$$dx(t) = a x(t) dt$$

$$\frac{1}{x(t)} dx(t) = a dt$$

$$\int \frac{1}{x(t)} dx(t) = \int a dt$$

$$\ln(x(t)) = at + c$$

$$x(t) = C \cdot e^{at}$$

$$\Rightarrow \boxed{x_{ic}(t) \quad x_0 e^{at}}$$

|| "dt"

|| "1/x(t)"

|| "

$$|| \text{III}: x(0) = x_0 \rightarrow C = x_0$$

· Insert into II: $y_{1c}(t) = cx_0 e^{at}$

ii) Forced Response:

· Problem setup:

$$\dot{x}(t) = ax(t) + bu(t) \quad (\text{IV})$$

$$y(t) = cx(t) + du(t) \quad (\text{V})$$

$$x(0) = 0 \quad (\text{VI})$$

· We will use a trick to solve IV:

1. Multiply both sides with e^{-at}

2. Integrate

$$\dot{x}(t) = ax(t) + bu(t)$$

|| Trick

|| I.B.P.

$$\int f' \cdot g dx = f \cdot g - \int f g' dx$$

$$\int_0^+ \underbrace{e^{-a\tau}}_{f'} \underbrace{x(\tau)}_g d\tau = \int_0^+ a e^{-a\tau} x(\tau) + b e^{-a\tau} u(\tau) d\tau$$

$$e^{-a\tau} x(\tau) \Big|_0^+ - \int_0^+ -a e^{-a\tau} x(\tau) d\tau =$$

$$\parallel \parallel x(0) = 0$$

$$e^{-at} x(t) + \int_0^+ a e^{-a\tau} x(\tau) d\tau = \int_0^+ a e^{-a\tau} x(\tau) + b e^{-a\tau} u(\tau) d\tau$$

$$\Rightarrow e^{-at} x(t) = \int_0^+ b e^{-a\tau} u(\tau) d\tau$$

$$x_F(t) = b \int_0^+ e^{a(t-\tau)} u(\tau) d\tau$$

$$\Rightarrow \text{Insert into (V)}: y_F(t) = cb \int_0^+ e^{a(t-\tau)} u(\tau) d\tau + du(t)$$

$\Rightarrow$ Due to linearity we have now have the solution to any LTI-system:

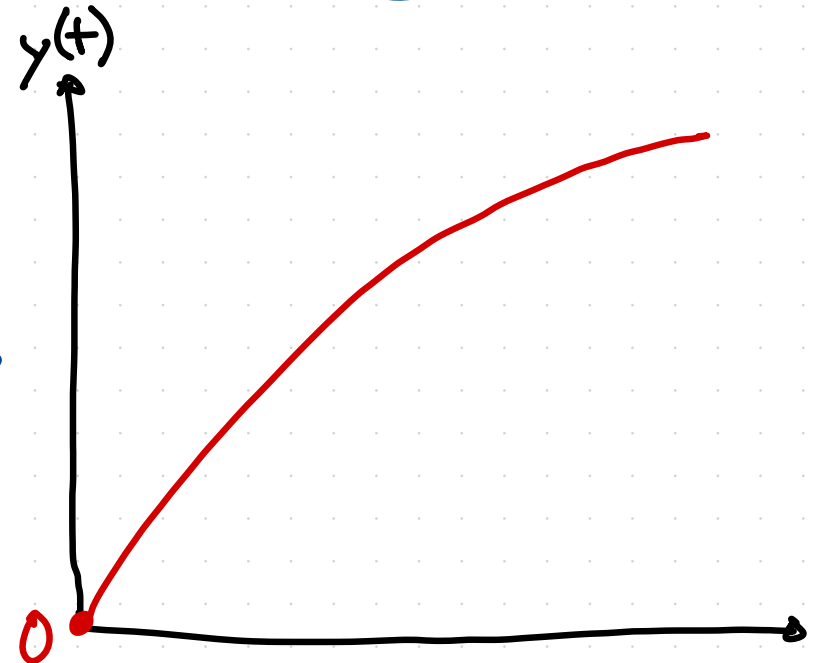
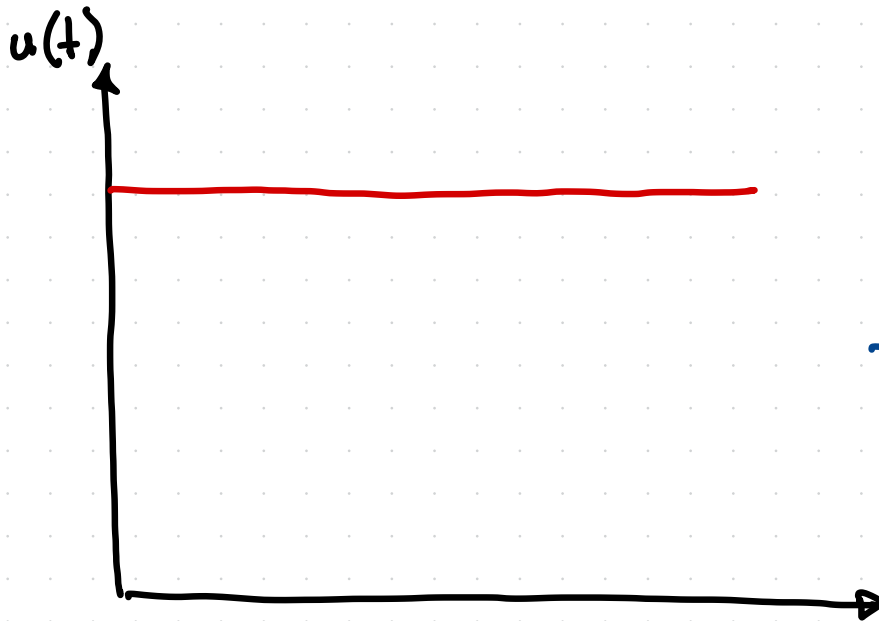
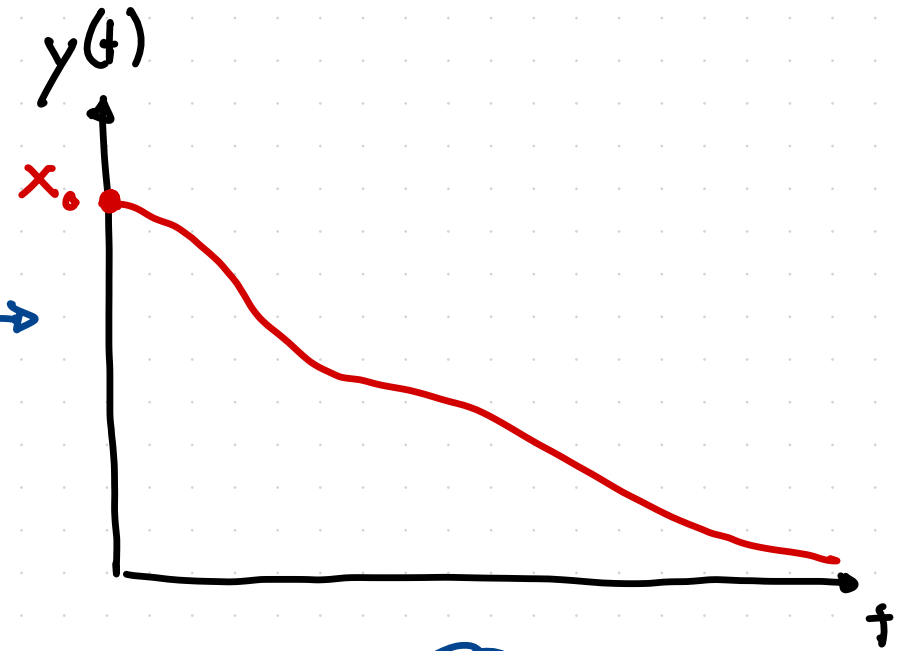
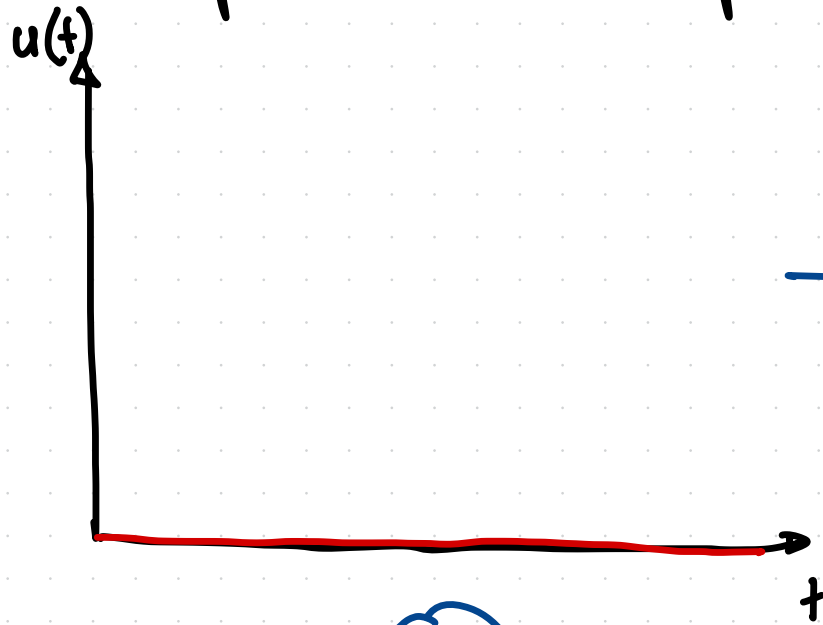
- $x(t) = x_{ic}(t) + x_F(t)$

$$x(t) = e^{at} x_0 + b \int_0^t e^{a(t-\tau)} u(\tau) d\tau$$

- $y(t) = y_{ic}(t) + y_F(t)$

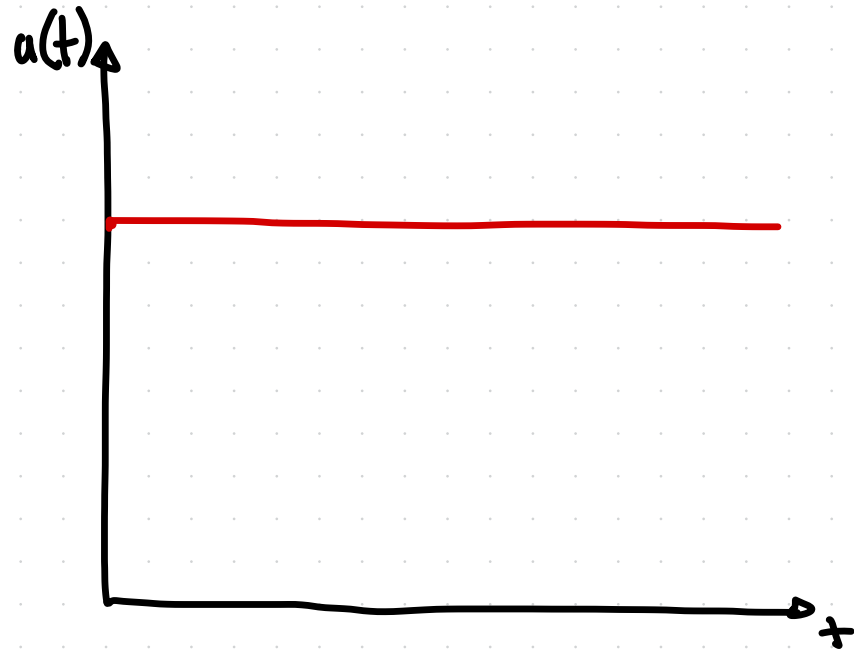
$$y(t) = ce^{at} x_0 + cb \int_0^t e^{a(t-\tau)} u(\tau) d\tau + du(t)$$

Graphical Interpretation:

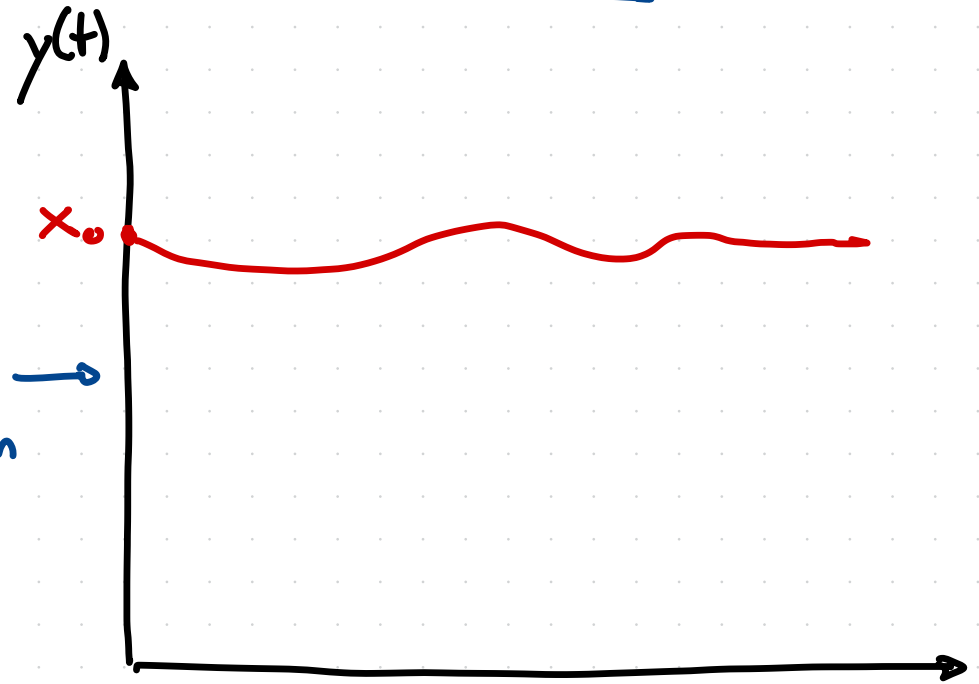


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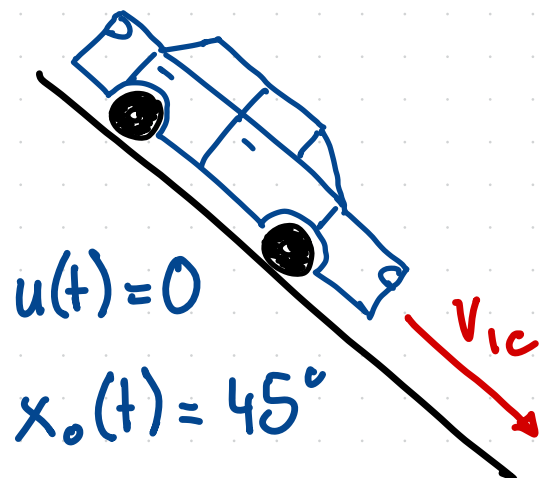
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$\rightarrow \Sigma_{lin} \rightarrow$



Intuition:



(+)



$u(t)=100\%$
 $x_0(t)=0$

(=)

$y(t) = y_{ic} + y_F$
 $= v_{ic} + v_c$
= CRAZY FAST!

Numerical example:

Given: $a = -1$ $x_0 = -1$

$$b = 1$$

$$c = 1$$

$$d = 0$$

$$u(t) = h(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \quad \dots \text{heaviside fct.}$$

Problem setup:

$$\dot{x}(t) = ax(t) + bu(t)$$

$$y(t) = cx(t) + du(t)$$

$$x(0) = x_0$$

$$\left. \begin{array}{l} \dot{x}(t) = -x(t) + u(t) \\ y(t) = x(t) \\ x(0) = -1 \end{array} \right\}$$

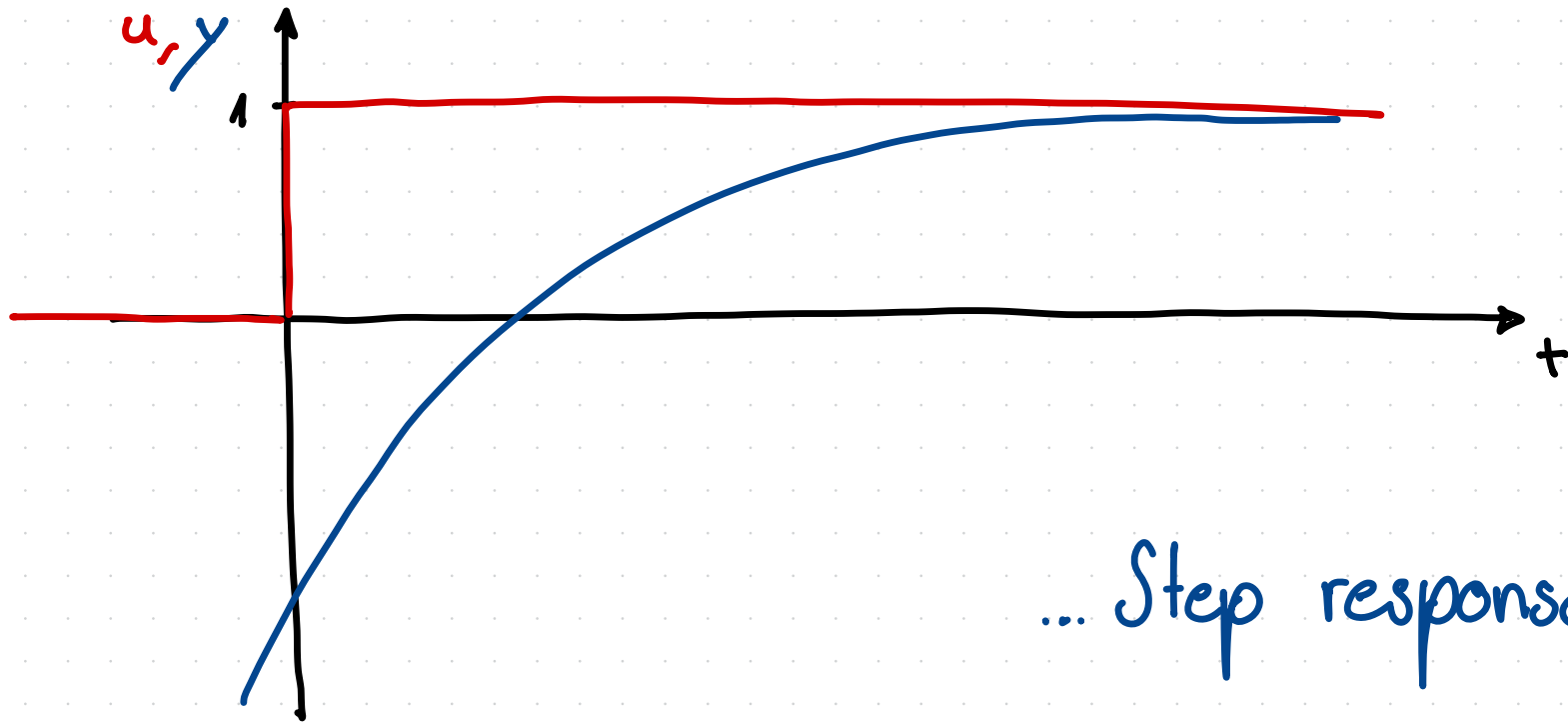
Formula for output:

$$y(t) = ce^{at}x_0 + cb \int_0^t e^{a(t-\tau)} u(\tau) d\tau + du(t)$$

⇒ Plug in:

$$y(t) = -e^{-t} + \int_0^t e^{-(t-\tau)} \overset{=1}{h(t)} d\tau = -e^{-t} + 1 - e^{-t}$$
$$= \underline{\underline{1 - 2e^{-2t}}}$$

· Graphically:



... Step response

Generalization:

• System matrices are in general non scalar!

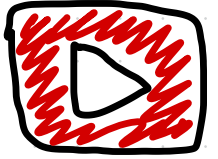
$$y(t) = c e^{at} x_0 + cb \int_0^t e^{a(t-\tau)} u(\tau) d\tau + du(t)$$



Watch the placement!

$$y(t) = C e^{At} x_0 + C \int_0^t e^{A(t-\tau)} B u(\tau) d\tau + D u(t)$$

→ How do we compute e^{At} ie. matrix exponentials?

↳ intuition: $YT \rightarrow 3\text{Blue}1\text{Brown}$, matrix exponentials 

• We compute matrix exponentials through Taylor series exp.:

$$T(e^x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \dots$$

↳ for $x = At$:

$$T(e^{At}) = I + At + \frac{1}{2} (At)^2 + \dots$$

→ possible, but computational effort might get out of hand.

Alternative:

• Theorem: If A is a diagonal matrix: $A = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

Eigenvalues

then:
$$e^{At} = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix}$$

What if our matrix is not diagonal?

Many matrices are diagonalizable

Def: A is diagonalizable, if $\exists$ a transformation matrix T , s.t.

$$D = T^{-1}AT$$

$$\hookrightarrow D = \text{diag}(\lambda_1, \dots, \lambda_i)$$

with λ_i ... Eigenvalues

$$\hookrightarrow T = (v_1, \dots, v_i)$$

with v_i ... Eigenvectors

$\hookrightarrow$ requires all eigenvectors to be linearly independent!

$\rightarrow$ How can we incorporate this into our system?

• Our system:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

... & we found T s.t. $\tilde{A} = T^{-1}AT$

→ What now?

1. Change coordinates: $x = T\tilde{x}$
2. Apply to system:

$$\left. \begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned} \right\} \begin{aligned}T\dot{\tilde{x}} &= AT\tilde{x} + Bu \\ y &= CT\tilde{x} + Du\end{aligned}$$

3. Simplify:

diagonalized A!

$$\dot{\tilde{x}} = (T^{-1}AT) \tilde{x} + (T^{-1}B)u$$

$$y = (CT) \tilde{x} + Du$$

4. Rename:

$$\cdot \tilde{A} := T^{-1}AT$$

$$\cdot \tilde{C} := CT$$

$$\cdot \tilde{B} = T^{-1}B$$

$$\cdot \tilde{D} = D$$

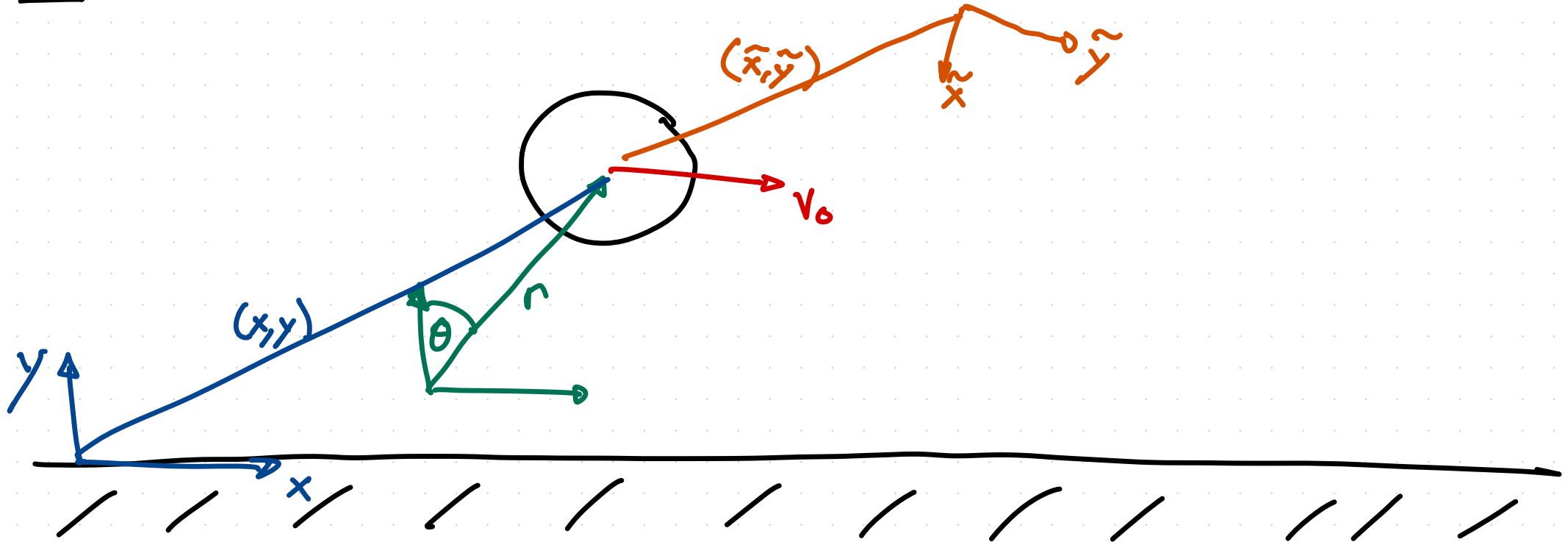
$\Rightarrow$

$$\begin{cases} \dot{\tilde{x}} = \tilde{A} \tilde{x} + \tilde{B} u \\ y = \tilde{C} \tilde{x} + \tilde{D} u \end{cases}$$

· Haven't we changed our system by changing A ?

→ No! There are infinitely many combinations of A, B, C, D that describe the same system & they can be transformed into each other through coordinate transformations

Ex.



Given:

$$\begin{aligned}\dot{\tilde{x}} &= \tilde{A} \tilde{x} + \tilde{B} u \\ y &= \tilde{C} \tilde{x} + \tilde{D} u\end{aligned}$$

... we can now easily evaluate $y(t)$:

$$y(t) = \tilde{C} e^{\tilde{A}t} x_0 + \tilde{C} \int_0^t e^{\tilde{A}(t-\tau)} \tilde{B} u(\tau) + \tilde{D} u(t)$$

What if our A is neither diagonal nor diagonalizable?

→ Bring it into Jordan form J

J is "almost diagonal":

$$J = \begin{bmatrix} J_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & J_i \end{bmatrix}$$

where:

Jordan block for EV λ_i :

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & 0 \\ \vdots & \lambda_i & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & \lambda_i \end{bmatrix}$$

Thm: If A is in Jordan form:

$$A = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} \Rightarrow e^{At} = \begin{bmatrix} e^{\lambda t} & t e^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix}$$

$\Rightarrow$ Allows us to again compute $y(t)$

- Computing e^{At} through the eigenvalues/-vectors of A is called modal decomposition

· Let's return to the initial condition response:

$$y_{ic}(t) = C e^{At} x_0 = C \exp\left(\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} t\right) x_0$$

$$= \begin{bmatrix} C_1 & C_2 \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} \begin{pmatrix} x_{01} \\ x_{02} \end{pmatrix}$$

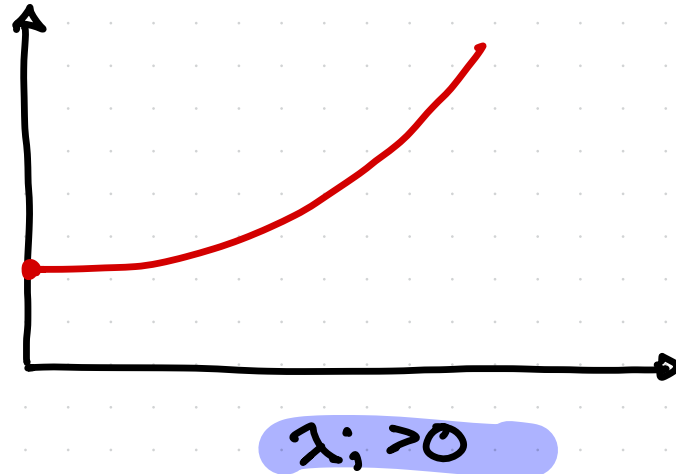
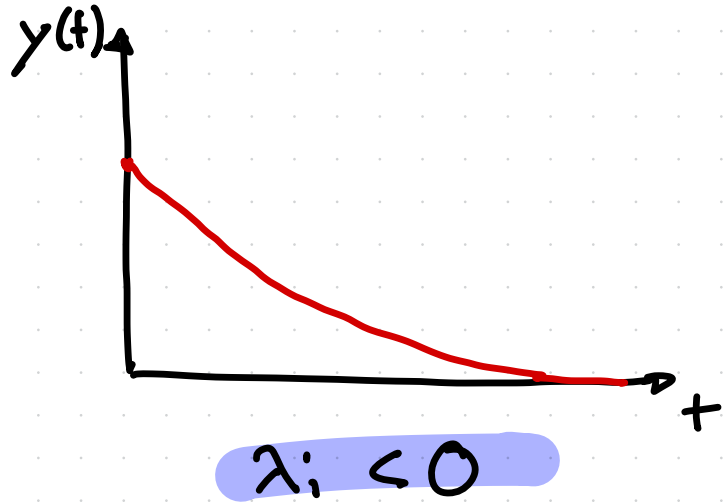
$$= C_1 e^{\lambda_1 t} \cdot x_{01} + C_2 e^{\lambda_2 t} x_{02} \quad \dots \text{"real" form} \rightarrow \lambda_i \in \mathbb{R}$$

· Considering the more general case for complex conjugate EV's:

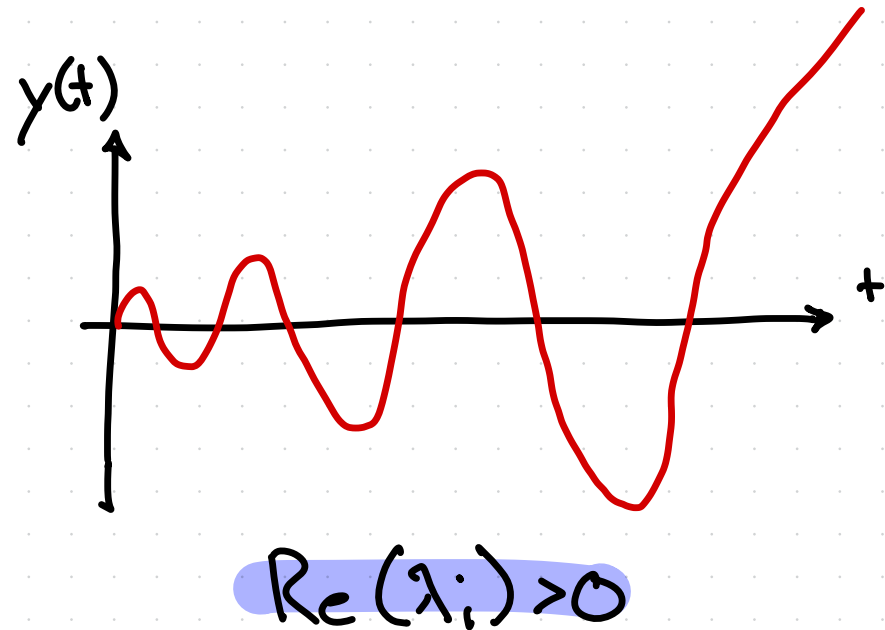
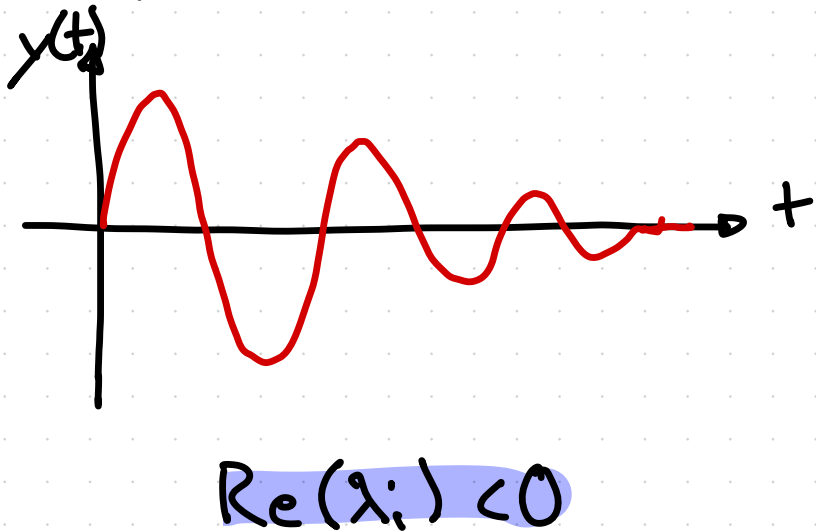
$$y_{ic}(t) = \alpha e^{\sigma t} \sin(\omega t + \phi) \quad \text{where } \lambda_i = \sigma \pm i\omega$$

Depending the EV's λ , $y(t)$ takes the following form:

i) $\lambda_i \in \mathbb{R}$:



ii) $\lambda_i \in \mathbb{C}$:



4. Stability

• We can classify 3 types of stability:

1) Lyapunov Stability:

Def: A system is Lyapunov stable, if for any bounded initial condition & zero input, the state remains bounded:

$$\|x_0\| < \varepsilon, \wedge u=0 \Rightarrow \|x(t)\| < \delta \quad \forall t \geq 0$$

$\Rightarrow$ Lyapunov stable if $\text{Re}(\lambda_i) \leq 0$

2) Asymptotic Stability:

Def: A system is asymptotically stable if, for any bounded initial condition & zero input, the state converges to zero:

$$\|x_0\| < \varepsilon \wedge u=0 \Rightarrow \lim_{t \rightarrow \infty} \|x(t)\| = 0$$

$\Rightarrow$ Asymptotically stable if $\operatorname{Re}(\lambda_i) < 0$

3) BIBO Stability:

Def: A system is Bounded - Input - Bounded - Output stable if for any bounded input, the output remains bounded:

$$\|u(t)\| < \varepsilon \quad \forall t \geq 0 \quad \wedge \quad x_0 = 0 \Rightarrow \|y(t)\| < \delta \quad \forall t \geq 0$$

... and when is a system unstable?

► A system is called **unstable** if not stable.

ETH zürich

Quiz:

Q1) All EV's of A must be >0 for the system to be unstable

☐ TRUE

☒ FALSE

Q2) An asymptotically stable system is always also Lyapunov stable

☒ TRUE

☐ FALSE

Q3) A system can have one real and one complex EV

☐ TRUE

☒ FALSE